

VLSI Design & Practice 03

22-bit RCA, SRCSA

Spring semester, 2026

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1. 22-bit Ripple Carry Adder Design

- Gate level description design

2. 22-bit Square Root Carry Select Adder Design

- Design the architecture of 22-bit Square Root Carry Select Adder

3. Simulation & Synthesis

- Check the functionality and synthesis results

4. Synthesis Results Analysis

- Compare the experimental results

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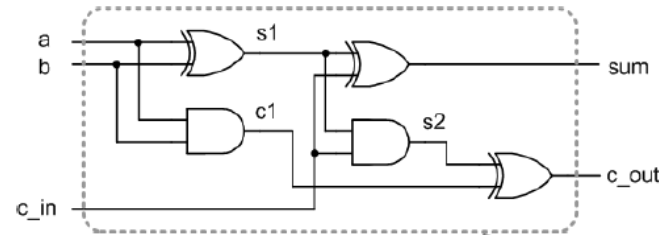
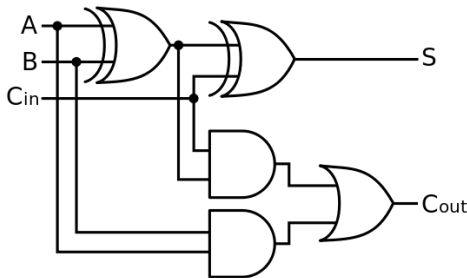
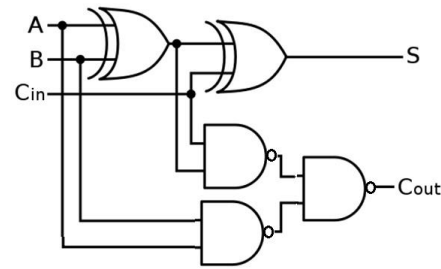
4. Synthesis Results Analysis

- Compare the experimental results

22-bit Ripple Carry Adder

- Basic Full Adder and Ripple Carry Adder

Gate-level adder architectures



Operating Conditions: nom_pvt Library: lec25dscc25_SS
Wire Load Model Mode: top

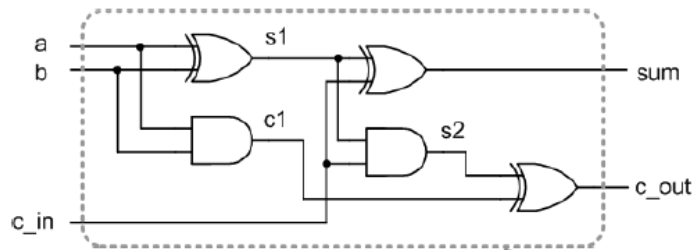
Startpoint: a[0] (input port)
Endpoint: sum[11] (output port)
Path Group: (none)
Path Type: max

report_timing

Point	Incr	Path
input external delay	0.00	0.00 r
a[0] (in)	0.00	0.00 r
da/d[0] (D_FF_12bit_0)	0.00	0.00 r
da/x0/d (D_FF_37)	0.00	0.00 r
da/x0/U5/Q (ib1s1)	0.07	0.07 f
da/x0/U1/Q (nnd2s2)	0.12	0.19 r
da/x0/_tmp54/Q (and3s2)	0.24	0.43 r
da/x0/U4/Q (aoi21s2)	0.14	0.56 f
da/x0/U3/Q (ib1s1)	0.20	0.76 r
da/x0/q (D_FF_37)	0.00	0.76 r
da/q[0] (D_FF_12bit_0)	0.00	0.76 r
fa_12bit/a[0] (fulladder_12bit)	0.00	0.76 r
fa_12bit/fa0/a (fulladder_1bit_0)	0.00	0.76 r
fa_12bit/fa0/U2/Q (xor2s1)	0.45	1.22 r
fa_12bit/fa0/U4/Q (aoi22s2)	0.26	1.48 f
fa_12bit/fa0/U3/Q (ib1s1)	0.18	1.66 r
fa_12bit/fa0/cout (fulladder_1bit_0)	0.00	1.66 r
fa_12bit/fa1/cin (fulladder_1bit_11)	0.00	1.66 r
fa_12bit/fa1/U4/Q (aoi22s2)	0.25	1.92 f
fa_12bit/fa1/U3/Q (ib1s1)	0.18	2.10 r
fa_12bit/fa1/cout (fulladder_1bit_11)	0.00	2.10 r
fa_12bit/fa2/cin (fulladder_1bit_10)	0.00	2.10 r
fa_12bit/fa2/U4/Q (aoi22s2)	0.25	2.35 f
fa_12bit/fa2/U3/Q (ib1s1)	0.18	2.53 r
fa_12bit/fa2/cout (fulladder_1bit_10)	0.00	2.53 r
fa_12bit/fa3/cin (fulladder_1bit_9)	0.00	2.53 r
fa_12bit/fa3/U4/Q (aoi22s2)	0.25	2.79 f
fa_12bit/fa3/U3/Q (ib1s1)	0.18	2.97 r
fa_12bit/fa3/cout (fulladder_1bit_9)	0.00	2.97 r
fa_12bit/fa4/cin (fulladder_1bit_8)	0.00	2.97 r
fa_12bit/fa4/U4/Q (aoi22s2)	0.25	3.22 f
fa_12bit/fa4/U3/Q (ib1s1)	0.18	3.41 r
fa_12bit/fa4/cout (fulladder_1bit_8)	0.00	3.41 r
fa_12bit/fa5/cin (fulladder_1bit_7)	0.00	3.41 r
fa_12bit/fa5/U4/Q (aoi22s2)	0.25	3.66 f
fa_12bit/fa5/U3/Q (ib1s1)	0.18	3.84 r
fa_12bit/fa5/cout (fulladder_1bit_7)	0.00	3.84 r

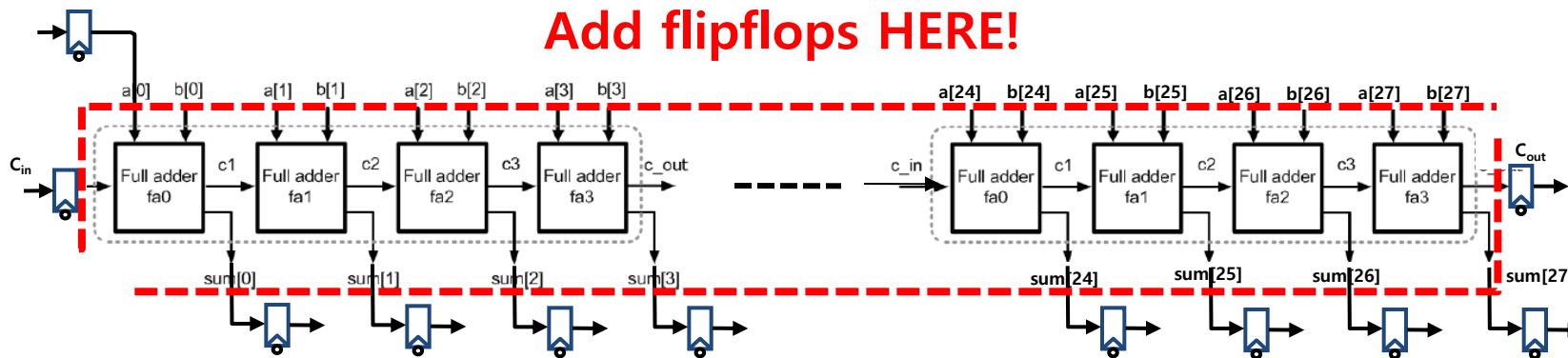
22-bit Ripple Carry Adder

- Add flipflops at input and output ports



Critical Path ?
In DC, is it VIOLATED ? Or MET ?

Reduce the clock until it is violated !
What is MAXIMUM Clock Frequency ?
What is AREA Change ?



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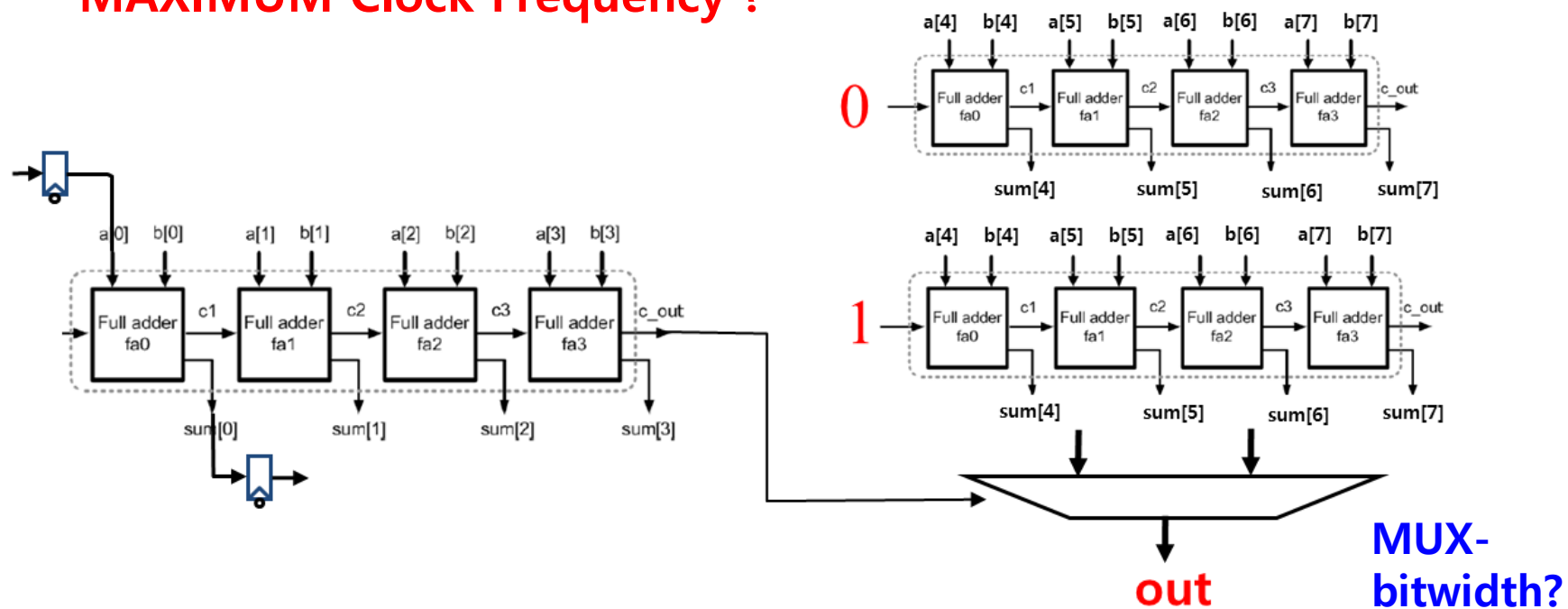
Square Root Carry Select Adder

- Why Carry Select Adder ?

Critical Path delay ?

Area change ?

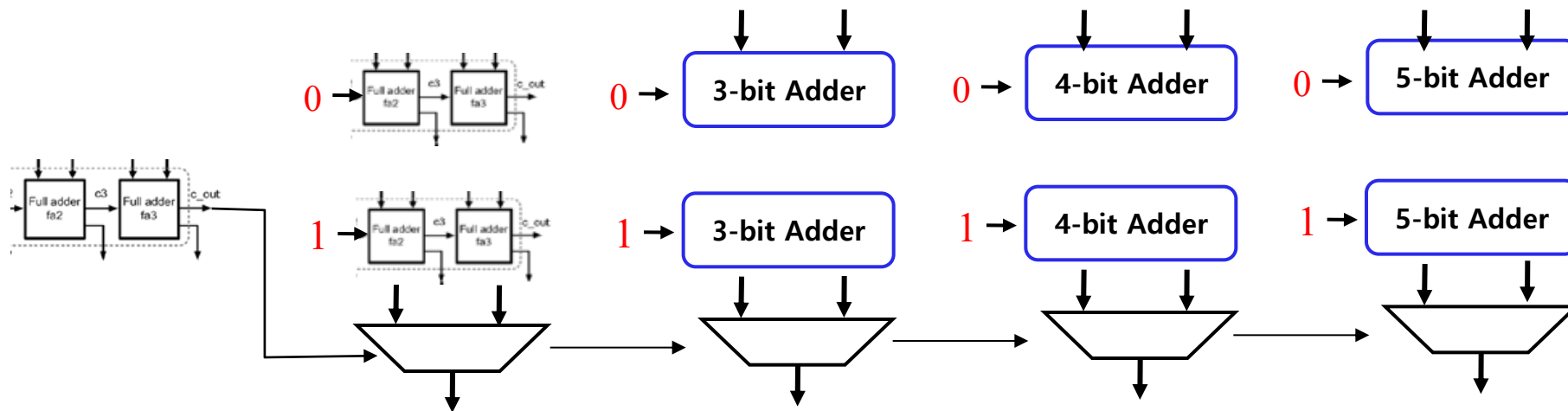
MAXIMUM Clock Frequency ?



Square Root Carry Select Adder

- 22-bit Square Root Carry Select Adder

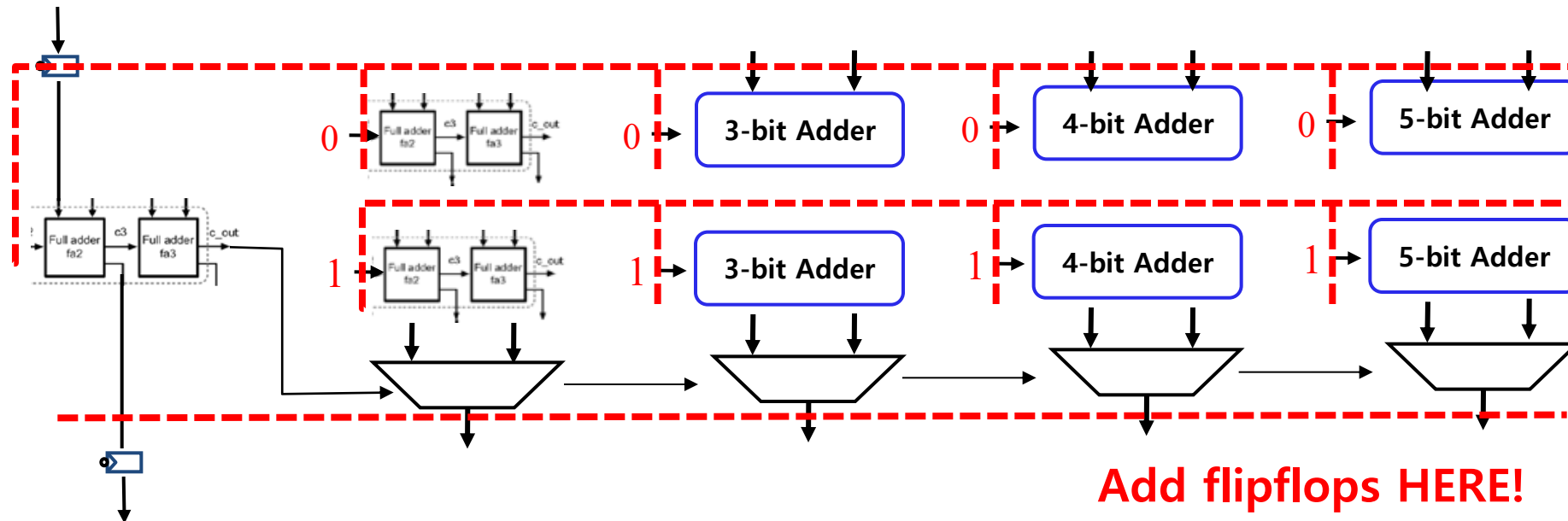
An Example of 16-bit Square Root Carry Select Adder



Decide the architecture of
22-bit Square Root Carry Select Adder
on your own !!

Square Root Carry Select Adder

- Add flipflops at all input and output ports



Reduce the clock until it is violated !
What is **MAXIMUM** Clock Frequency ?
What is **AREA** Change ?

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3. **Simulation & Synthesis**

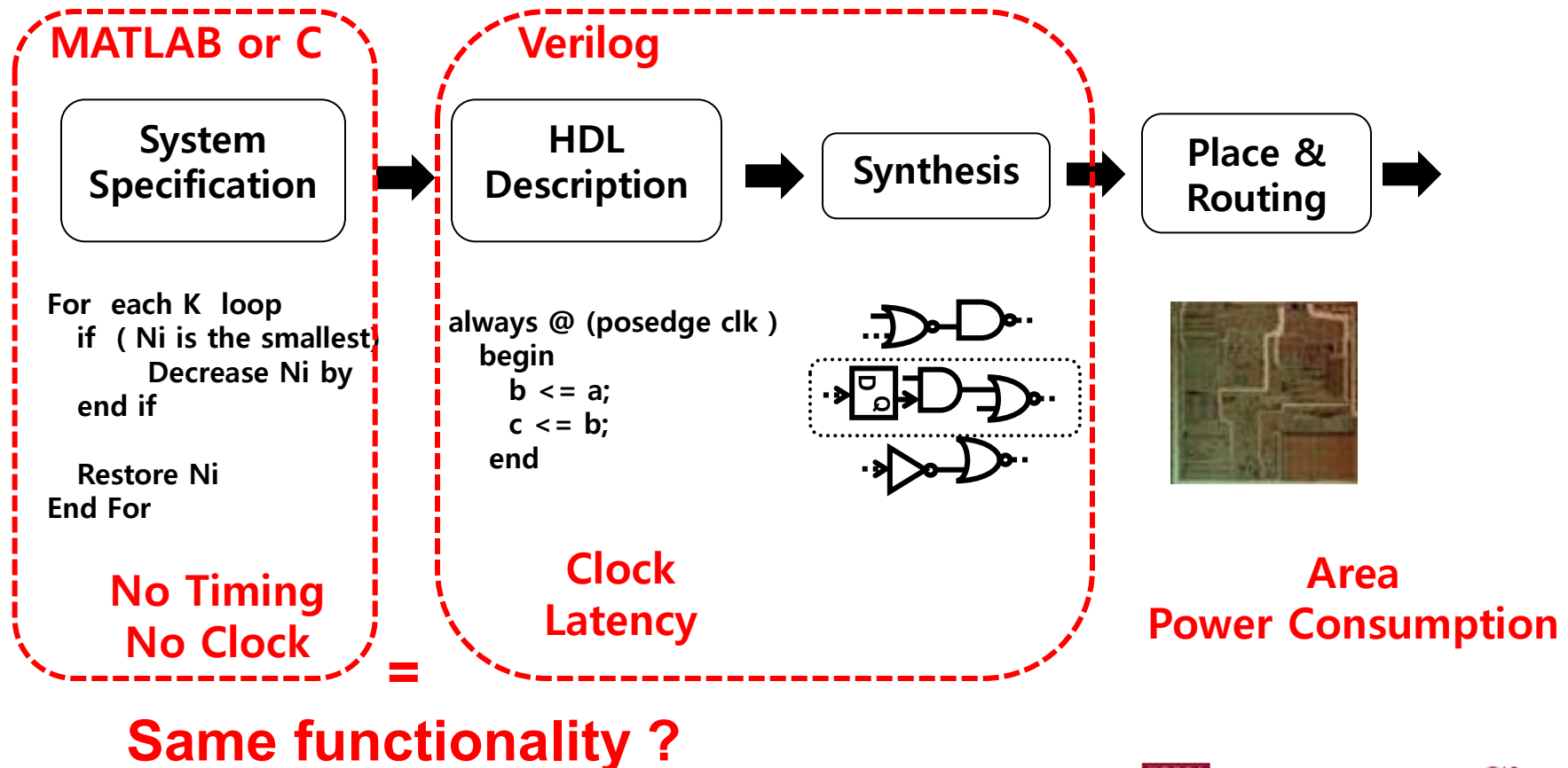
- **Check the functionality and Synthesis results**

4. Synthesis Results Analysis

- Compare the experimental results

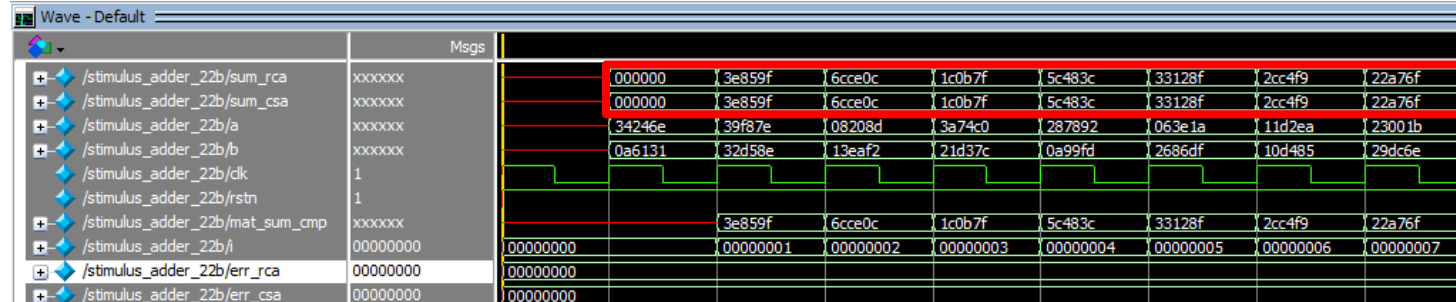
Functional Verifications

How do we verify the functionality of the design ?

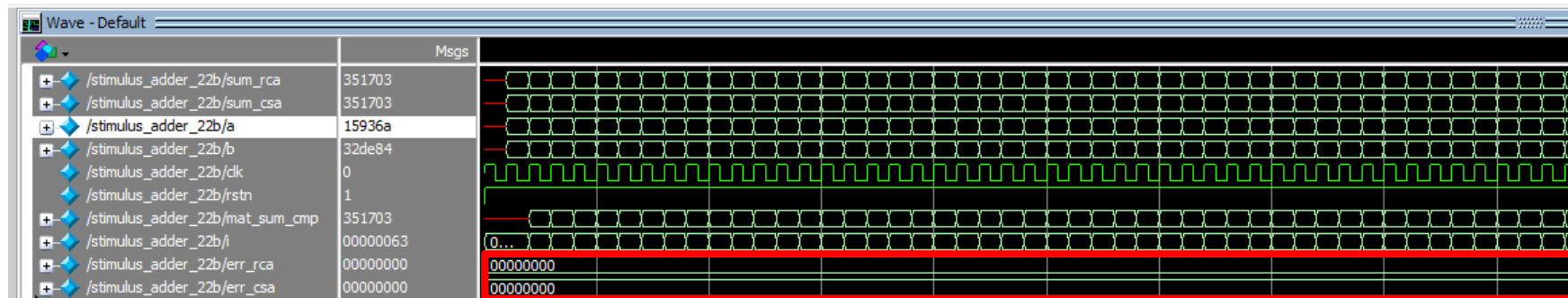


Functional Verifications

- Simulation results



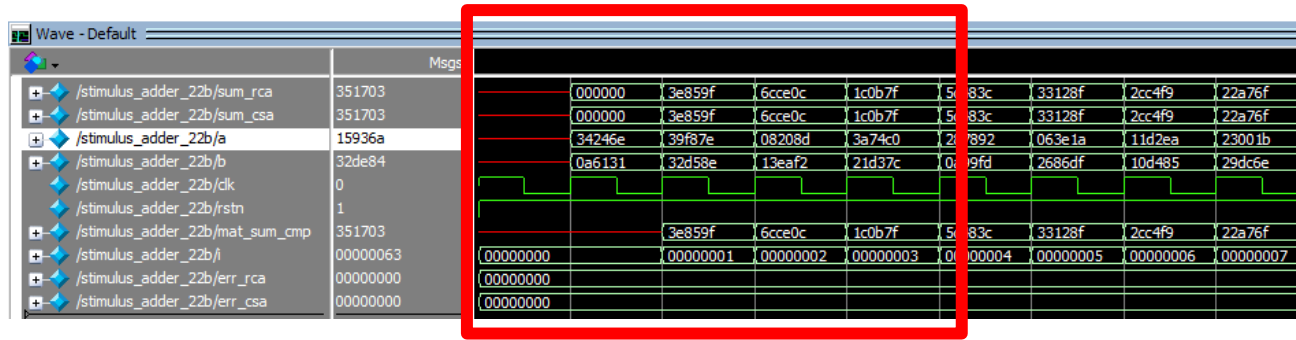
- Check output of **two adders** and Matlab results are '**same**'



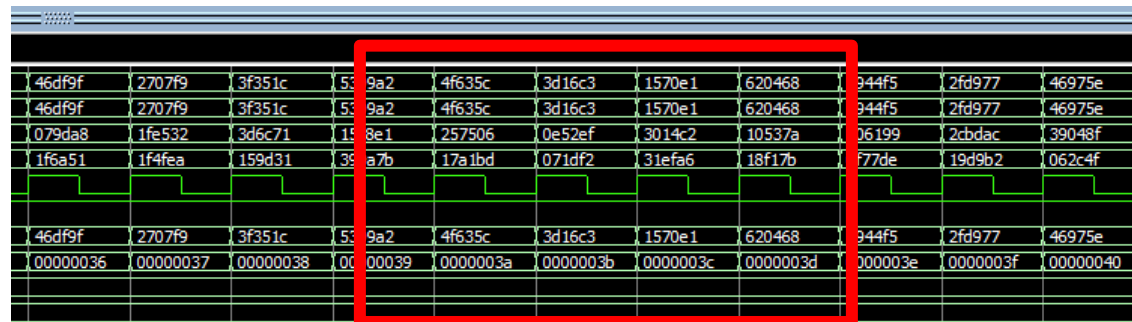
- Check the value of error counting number equals '**zero**'

Tip) How to report waveform?

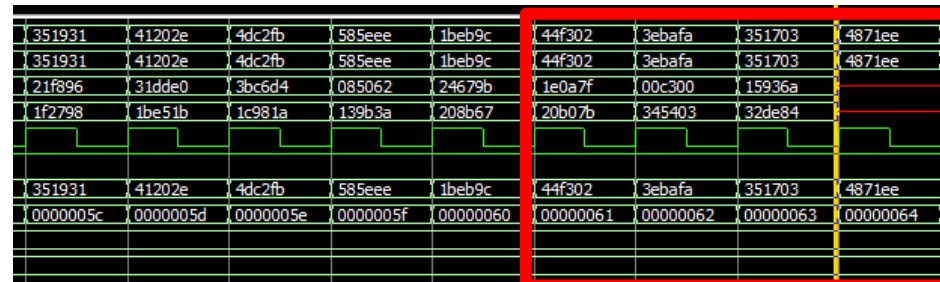
Show the start, middle and end!



Show correct
initialization



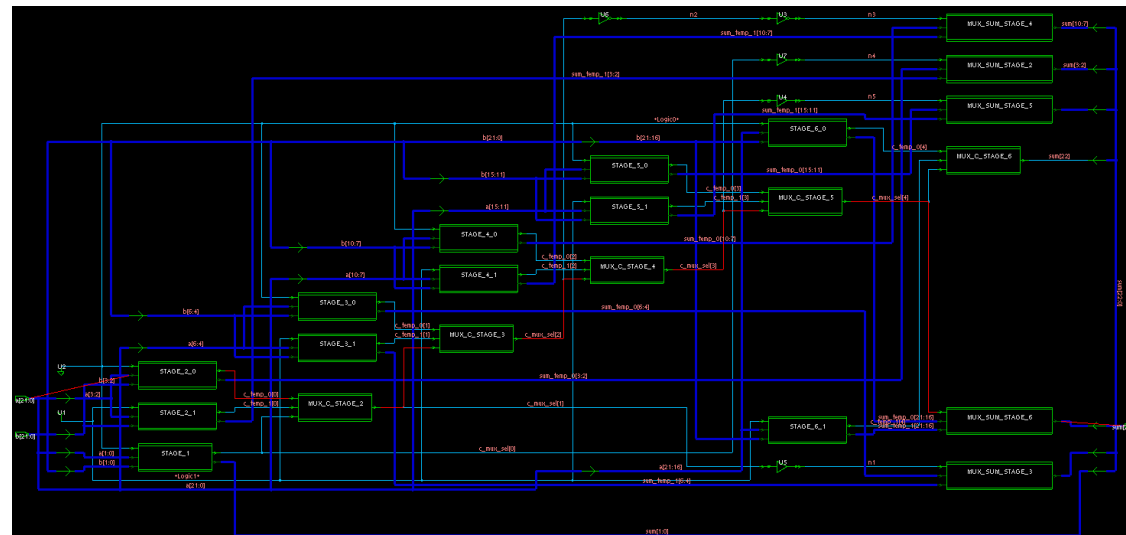
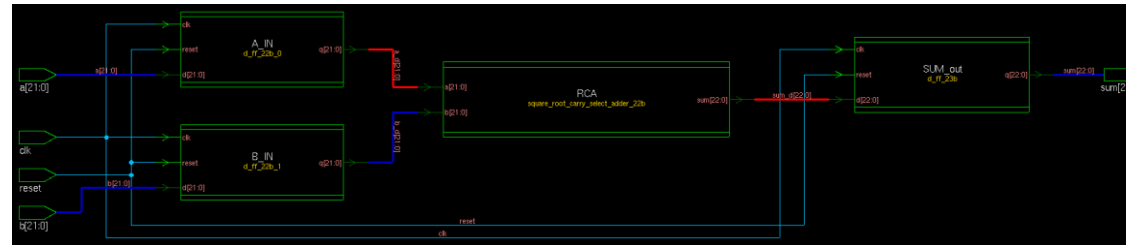
Show no errors
during operation



Show no errors
after all tests

Find the maximum clock frequency

- **Synthesis results** 22-bit Ripple Carry Adder & Square Root Carry Select Adder
- Check the maximum clock speed and critical path



Ctrl+H
to highlight
critical path

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Check results and Submit report

- **Submit result report**
 - Show the **timing diagram** and **synthesis results**
 - Don't include the Verilog code in the report
 - Attach the **zip file** including all the related .v files
- **Check the **synthesis results** for**
 - (1) 22-bit Ripple Carry Adder
 - (2) 22-bit Square Root Carry Select Adder
 - Add flipflops at the input and output
 - Check the highlighted critical path, area report and timing report
 - Compare area and critical path delay between (1) and (2)

Check results and Submit report

- Due : Before the next class to your group TA
(담당 조교에게 제출 바랍니다)

Title of the Mail : 2026_VLSI_Practice03_학번_이름

Report : Report_Practice03_학번_이름.pdf

Zip file : Verilog_Practice03_학번_이름.zip